Inverse Transform Watermark

Remark 1.

Let X be a random variable with CDF $F_X(x) = \mathbb{P}(X \leq x)$.
Let $U \sim U(0,1)$.
Let $Y = F_X^{-1}(U)$, where $F^{-1}(p) = \inf\{x \in \mathbb{R} : F(x) \geq p\}, \quad \forall p \in [0,1]$.
Then $Y \sim X$.

We see "unbiasedness" there.

How does it work?

Imposing an Order on the Vocabulary

The Inverse Transform method requires a 1D real number line to define a "Cumulative" function (i.e., you must be able to say $X \leq x$).

However, the vocabulary $\mathcal{W}$ is a set of words (categorical data) with no inherent mathematical order.

To solve this, we introduce a permutation $\pi : \mathcal{W} \to \{1, 2, \ldots, |\mathcal{W}|\}$.

- $\pi(w)$ assigns a unique integer rank to a token w.
- $\pi^{-1}(i)$ takes rank i and returns the corresponding token. This permutation π allows us to arrange the discrete probabilities of the NTP distribution $\boldsymbol{P}$ into a 1D sequence on the integer line. The probability of the token at position i is $P_{\pi^{-1}(i)}$.

Remark 2.

Why need permutation?
Even if we use some fixed ordering, inverse transform sampling still works and the watermark is still unbiased (intuitively, only the "length" but not the position matters).
But we need this permutation to preserve security (if use some fixed order, then w_t and the hidden pseudorandom number U_t's coupling relationship are static and clear.) and more importantly, validity of the pivotal quantity under H_0.

Deriving the CDF $F(x; \pi)$

Consider the multinomial distribution with probability mass $P_{\pi^{-1}(i)}$ at i for $1 \leq i \leq |\mathcal{W}|$.

Now that the tokens are ordered from 1 to $|\mathcal{W}|$, we can define the CDF. The CDF $F(x; \pi)$ is the cumulative sum of probabilities for all tokens assigned a rank less than or equal to x.

For any real number x: $F(x;\pi) = \sum_{i=1}^{\lfloor x \rfloor} P_{\pi^{-1}(i)}$

To express this in terms of the original vocabulary $\mathcal{W}$ rather than the integer indices, we sum over all tokens $w' \in \mathcal{W}$, but use an indicator function $\mathbf{1}_{\{\pi(w') \le x\}}$ to only include the probabilities of tokens whose rank is less than or equal to x.
$F(x;\pi) = \sum_{w' \in \mathcal{W}} P_{w'} \cdot \mathbf{1}_{\{\pi(w') \le x\}}$

Deriving the Generalized Inverse $F^{-1}(U;\pi)$

Because the distribution is discrete, the CDF $F(x;\pi)$ is a step function. we use the generalized inverse. For a uniform random variable $U \sim U(0,1)$,
$F^{-1}(U;\pi) = \min\{i \in \mathbb{Z} : F(i;\pi) \ge U\}$

Which is $F^{-1}(U;\pi) = \min\left\{i : \sum_{w' \in \mathcal{W}} P_{w'} \cdot \mathbf{1}_{\{\pi(w') \le i\}} \ge U\right\}$

Deriving the Decoder $\mathcal{S}^{\mathrm{inv}}$

$w_t = \mathcal{S}^{\mathrm{inv}}(\boldsymbol{P}, \zeta) := \pi^{-1}(F^{-1}(U;\pi))$ where $\zeta = (\pi, U)$ and the permutation π is uniformly at random.

We show this watermark is unbiased.

We need to show $\mathbb{P}(\mathcal{S}^{\mathrm{inv}}(\boldsymbol{P}, \zeta) = w) = P_w$ for all $w \in \mathcal{W}$.

Proof.
Let w be an arbitrary token in $\mathcal{W}$. Let its rank under the permutation be $k = \pi(w)$. Applying π to both sides of $\pi^{-1}(F^{-1}(U;\pi)) = \mathcal{S}^{\mathrm{inv}}(\boldsymbol{P}, \zeta) = w$, equivalently $F^{-1}(U;\pi) = \pi(w) = k$.

And so equivalently $F(k-1;\pi) < U \le F(k;\pi)$.

Since $F(k;\pi) = \sum_{w' \in \mathcal{W}} P_{w'} \cdot \mathbf{1}_{\{\pi(w') \le k\}}$ and $F(k-1;\pi) = \sum_{w' \in \mathcal{W}} P_{w'} \cdot \mathbf{1}_{\{\pi(w') < k\}}$, then equivalently

$$\sum_{w' \in \mathcal{W}} P_{w'} \cdot \mathbf{1}_{\{\pi(w') < k\}} < U \le \sum_{w' \in \mathcal{W}} P_{w'} \cdot \mathbf{1}_{\{\pi(w') \le k\}}$$

The length of this interval is $F(k;\pi) - F(k-1;\pi) = P_{\pi^{-1}(k)} = P_w$.

Since U is uniformly distributed on the interval $(0,1)$, then
$\mathbb{P}(\mathcal{S}^{\mathrm{inv}}(\boldsymbol{P}, \zeta) = w) = \mathbb{P}(F(k-1;\pi) < U \le F(k;\pi)) = P_w$.

Or we can use the first remark in this document.
$\mathbb{P}(X = i) = P_{\pi^{-1}(i)}$, if $Y = F_X^{-1}(U)$, then $Y \sim X$.
$\mathbb{P}(Y = i) = \mathbb{P}(X = i) = P_{\pi^{-1}(i)}$.
Note $\mathbb{P}(\text{output } w) = \mathbb{P}(Y = \pi(w))$, thus $\mathbb{P}(Y = \pi(w)) = \mathbb{P}(X = \pi(w)) = P_{\pi^{-1}(\pi(w))} = P_w$.

□

Main results

By "under H_1, in contrast, a larger U_t value suggests a larger value of $\pi_t(w_t)$ in distribution for the watermarked text, which can be gleaned from (23)", it means that: recall from above, we know if say $\pi_t(w) = k$, then token w is chosen iff $F(k-1;\pi_t) < U_t \leq F(k;\pi_t)$.

Typo above eqaution (37). Should be $\pi(l) = w$ not $\pi(l) = k$ in the index of summation.

Proof of Lemma 4.1

For H_0, From

$$\mathbb{P}(Y_t^{\text{dif}} \leq r \mid H_0) = \sum_{w \in \mathcal{W}} \mathbb{P}\left(U_t \in [\eta(w) - r, \eta(w) + r] \cap [0,1], \pi_t(w_t) = w \mid H_0\right)$$

Use results from lemma C.1 for H_0: the joint probability is $\frac{1}{|\mathcal{W}|}$ multiplied by the probability of U_t falling into the target interval. Because $U_t \sim U(0,1)$, the probability of U_t falling into a specific interval within $[0,1]$ is exactly the length of that interval.

For H_1, note changed order of summation and replaced probability to uniform measure.